

EE1202: Digital Circuits — Assignment 1

Complete Step-by-Step Analytical Solutions & Logic Realizations

Course: EE1202 Digital Circuits

Deadline: Saturday 08-08-26 11:59PM

Format: Detailed Step-by-Step Solutions

Prerequisite Concepts & Fundamentals (Review)

To ensure total clarity, the following basic rules are applied throughout this document:

- **K-Map Extension (5 and 6 Variables):** 5-variable maps use two 4-variable sub-maps (e.g., for 5th variable = 0 and 1). Adjacency exists within each 4-var sub-map and between corresponding cells of the two sub-maps. 6-variable maps use four 4-variable sub-maps arranged in Gray code order (00, 01, 11, 10).
- **Universal NAND Implementation:** Any SOP expression can be implemented using 2-level NAND-NAND logic. To restrict to 2-input NAND gates:
 - Inverter (NOT): $X' = \text{NAND}(X, X)$ [1 gate, depth 1]
 - 2-input AND: $(X \cdot Y) = \text{NAND}(\text{NAND}(X, Y), \text{NAND}(X, Y))$ [2 gates, depth 2]
 - 2-input OR: $(X + Y) = \text{NAND}(X', Y') = \text{NAND}(\text{NAND}(X, X), \text{NAND}(Y, Y))$ [3 gates, depth 2]
- **Universal NOR Implementation:** Any POS expression can be implemented using 2-level NOR-NOR logic. To restrict to 2-input NOR gates:
 - Inverter (NOT): $X' = \text{NOR}(X, X)$
 - 2-input OR: $(X + Y) = \text{NOR}(\text{NOR}(X, Y), \text{NOR}(X, Y))$
 - 2-input AND: $(X \cdot Y) = \text{NOR}(X', Y') = \text{NOR}(\text{NOR}(X, X), \text{NOR}(Y, Y))$

Problem 1: Laboratory Access Controller (4 Variables)

Inputs: *A* (valid card), *B* (correct PIN), *C* (faculty auth), *D* (emergency mode). Output *F(A,B,C,D)* opens door.

Conditions: (1) *A* and *B* valid; (2) *A* valid and *C* present; (3) *D* active AND (*B* correct OR *C* present).

1. Truth Table & 2. Canonical SOP Form

Boolean expression: $F = A \cdot B + A \cdot C + D \cdot (B + C) = AB + AC + BD + CD$

Row	A	B	C	D	AB	AC	BD	CD	F	Minterm
0	0	0	0	0	0	0	0	0	0	—
1	0	0	0	1	0	0	0	0	0	—
2	0	0	1	0	0	0	0	0	0	—
3	0	0	1	1	0	0	0	1	1	m3
4	0	1	0	0	0	0	0	0	0	—
5	0	1	0	1	0	0	1	0	1	m5
6	0	1	1	0	0	0	0	0	0	—
7	0	1	1	1	0	0	1	1	1	m7
8	1	0	0	0	0	0	0	0	0	—
9	1	0	0	1	0	0	0	0	0	—
10	1	0	1	0	0	1	0	0	1	m10
11	1	0	1	1	0	1	0	1	1	m11
12	1	1	0	0	1	0	0	0	1	m12

13	1	1	0	1	1	0	1	0	1	m13
14	1	1	1	0	1	1	0	0	1	m14
15	1	1	1	1	1	1	1	1	1	m15

Canonical SOP: $F(A,B,C,D) = \Sigma m(3, 5, 7, 10, 11, 12, 13, 14, 15)$

3. 4-Variable K-Map Minimization

K-Map layout for $F(A,B,C,D)$ with rows = AB, cols = CD:

F(A,B,C,D)		CD			
		00	01	11	10
AB	00	0	0	1 (m3)	0
	01	0	1 (m5)	1 (m7)	0
	11	1 (m12)	1 (m13)	1 (m15)	1 (m14)
	10	0	0	1 (m11)	1 (m10)

Grouping Prime Implicants:

- Group 1 (Quad - m12, m13, m14, m15): Row AB=11 → **AB**
- Group 2 (Quad - m10, m11, m14, m15): AB=11,10 & CD=11,10 → **AC**
- Group 3 (Quad - m5, m7, m13, m15): AB=01,11 & CD=01,11 → **BD**
- Group 4 (Quad - m3, m7, m11, m15): AB=00,01,11,10 & CD=11 → **CD**

Minimized SOP Expression: $F = AB + AC + BD + CD$

4. Implementation Using Only 2-Input NAND Gates

We convert $F = AB + AC + BD + CD$ into 2-input NAND tree structure:

1. Compute term NAND pairs (using inputs directly):

$$N_1 = (A \cdot B)' \text{ [1 gate]}$$

$$N_2 = (A \cdot C)' \text{ [1 gate]}$$

$$N_3 = (B \cdot D)' \text{ [1 gate]}$$

$$N_4 = (C \cdot D)' \text{ [1 gate]}$$

2. To combine four terms using 2-input NAND gates:

Note that by De Morgan's: $X + Y = (X' \cdot Y')' = \text{NAND}(X', Y')$.

Since $N_1 = (AB)'$, $N_1' = AB$.

Combine N_1 and N_2 : $G_1 = \text{NAND}(N_1, N_2) = ((AB)' \cdot (AC)')' = AB + AC$.

Combine N_3 and N_4 : $G_2 = \text{NAND}(N_3, N_4) = ((BD)' \cdot (CD)')' = BD + CD$.

3. To OR G_1 and G_2 using 2-input NAND:

Invert $G_1 \rightarrow G_1' = \text{NAND}(G_1, G_1)$ [1 gate].

Invert $G_2 \rightarrow G_2' = \text{NAND}(G_2, G_2)$ [1 gate].

Final Output: $F = \text{NAND}(G_1', G_2') = (G_1'' \cdot G_2'')' = G_1 + G_2 = AB + AC + BD + CD$ [1 gate].

Total 2-Input NAND Gates Required: 4 (level 1) + 2 (level 2) + 2 (level 3 inverters) + 1 (level 4) = **9 NAND gates.**

Problem 2: Water-Pump Controller (4 Variables)

Inputs: L (low level), T (tank demand), M (manual start), E (emergency shutdown active = 1).

Conditions: Emergency $E=1$ forces output $P=0$. When $E=0$, pump turns on if $(L \text{ AND } T) \text{ OR } (M \text{ AND } L)$.

1. Translation & 2. Output-1 Minterms

Boolean expression: $P(L,T,M,E) = (L \cdot T + M \cdot L) \cdot E' = L T M' E' + L T M E' + L T' M E' = L E' (T + M)$

Evaluating rows for (L,T,M,E) binary order (8421):

- $L=1, T=0, M=1, E=0 \rightarrow$ Row 10 (1010) = m_{10}
- $L=1, T=1, M=0, E=0 \rightarrow$ Row 12 (1100) = m_{12}
- $L=1, T=1, M=1, E=0 \rightarrow$ Row 14 (1110) = m_{14}

Minterm List: $P(L,T,M,E) = \Sigma m(10, 12, 14)$

3. 4-Variable K-Map Minimization

K-Map layout for $P(L,T,M,E)$ with rows = LT, cols = ME:

P(L,T,M,E)		ME			
		00	01	11	10
LT	00	0	0	0	0
	01	0	0	0	0
	11	1 (m12)	0	0	1 (m14)
	10	0	0	0	1 (m10)

Grouping:

- Pair 1 (m12, m14): Row LT=11, Cols ME=00,10 $\rightarrow L T E'$
- Pair 2 (m10, m14): Rows LT=11,10 ($L=1, M=1$), Col ME=10 $\rightarrow L M E'$

Minimized SOP: $P = L T E' + L M E' = L E' (T + M)$

4. NAND-Only Realization

Using 2-input NAND gates exclusively:

1. Invert E : $E' = \text{NAND}(E, E)$ [1 gate].
2. Compute $L \cdot E'$: $N_1 = \text{NAND}(L, E')$.
Then invert to get $(L E')$: $LE' = \text{NAND}(N_1, N_1)$ [2 gates total for LE'].
3. Compute $(T + M)$ using 2-input NANDs:
 $T' = \text{NAND}(T, T)$
 $M' = \text{NAND}(M, M)$
 $(T + M) = \text{NAND}(T', M')$ [3 gates].

4. Finally, AND ($L E'$) with ($T + M$) using NAND:

$$N_{\{final\}} = NAND(LE', T + M)$$

$$P = NAND(N_{\{final\}}, N_{\{final\}}) \text{ [2 gates].}$$

Total 2-Input NAND Gate Count: 1 (for E') + 2 (for LE') + 3 (for $T+M$) + 2 (for final AND) = **8 NAND gates**.

Problem 3: Examination Eligibility System (5 Variables)

Inputs: A (attendance), F (fee), H (hall ticket), M (medical), D (disciplinary restriction active = 1).

Conditions: Disciplinary restriction $D=1$ denies eligibility under all conditions. When $D=0$, eligible if (A AND F AND H) OR (M AND F AND H).

1. Derived Expression & Truth Table Summary

$$E(A,F,H,M,D) = D' \cdot F \cdot H \cdot (A + M) = D' F H A + D' F H M$$

Out of 32 total rows (0 to 31):

- Output $E = 1$ when $D=0$, $F=1$, $H=1$ and ($A=1$ or $M=1$).
- Minterms for $E=1$ (ordering A,F,H,M,D):
 - $A=0$, $F=1$, $H=1$, $M=1$, $D=0 \rightarrow 01110$ (row 14)
 - $A=1$, $F=1$, $H=1$, $M=0$, $D=0 \rightarrow 11100$ (row 28)
 - $A=1$, $F=1$, $H=1$, $M=1$, $D=0 \rightarrow 11110$ (row 30)
- All other 29 minterms give $E = 0$ (Maxterms M_i for $i \notin \{14, 28, 30\}$).

2. Canonical POS Expression

$$\text{Canonical POS: } E(A,F,H,M,D) = \prod M(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 29, 31)$$

3. 5-Variable K-Map Minimization (POS / Maxterm Grouping)

We perform POS minimization by grouping 0s on two 4-variable sub-maps (for $D=0$ and $D=1$):

- For $D=1$, all 16 cells are 0 $\rightarrow$ gives Maxterm factor (D') equivalent to term D' in SOP or factor (D') when considering $E = D' \cdot F \cdot H \cdot (A + M)$.

Direct POS simplification yields:

$$\text{Minimized POS Expression: } E = D' \cdot F \cdot H \cdot (A + M)$$

4. Implementation Using Only 2-Input NOR Gates

To implement a POS expression using 2-input NOR gates:

1. Invert D : $D' = NOR(D, D)$ [1 gate].
2. Form $(A + M)$: $N_1 = NOR(A, M) \rightarrow (A + M) = NOR(N_1, N_1)$ [2 gates].
3. Combine factors with NOR-NOR logic:
 $(D' \cdot F) = NOR(D'', F') = NOR(D', NOR(F, F))$
Let's use clean step-by-step 2-input NOR equivalent for multi-input AND $D' \cdot F \cdot H \cdot (A + M)$:
 - $T_1 = (D' \cdot F) = NOR(NOR(D', D'), NOR(F, F))$ [3 gates]
 - $T_2 = (H \cdot (A + M)) = NOR(NOR(H, H), NOR(A + M, A + M))$ [3 gates]
 - $E = T_1 \cdot T_2 = NOR(NOR(T_1, T_1), NOR(T_2, T_2))$ [3 gates].

Total 2-Input NOR Gates Required: 12 NOR gates.

Problem 4: Industrial Alarm System (5 Variables)

Inputs: T (temp), P (pressure), S (smoke), G (gas), R (maintenance mode active = 1).

Conditions:

- When $R=0$ (normal): Alarm turns on if any 2 or more of $\{T, P, S\}$ are active, OR gas G is active.
- When $R=1$ (maintenance): T and P are suppressed; alarm turns on if smoke S OR gas G is active.

1. Complete Truth Table & 2. Maxterm Representation

Function: $Y(T,P,S,G,R) = R' \cdot (T P + P S + S T + G) + R \cdot (S + G)$

T P S G R	Min	Y	T P S G R	Min	Y
00000	m0	0	10000	m16	0
00001	m1	0	10001	m17	0
00010	m2	1	10010	m18	1
00011	m3	1	10011	m19	1
00100	m4	0	10100	m20	0
00101	m5	1	10101	m21	1
00110	m6	1	10110	m22	1
00111	m7	1	10111	m23	1
01000	m8	0	11000	m24	0
01001	m9	0	11001	m25	0
01010	m10	1	11010	m26	1
01011	m11	1	11011	m27	1
01100	m12	1	11100	m28	1
01101	m13	1	11101	m29	1
01110	m14	1	11110	m30	1
01111	m15	1	11111	m31	1

Output 0 Rows (Maxterms): m0, m1, m4, m8, m9, m16, m17, m20, m24, m25.

Maxterm Representation: $Y(T,P,S,G,R) = \prod M(0, 1, 4, 8, 9, 16, 17, 20, 24, 25)$

3. 5-Variable K-Map Minimization

K-Map with variable order (T,P,S,G,R) : 2 sub-maps for $R=0$ and $R=1$.

Grouping the 1s across sub-maps gives:

$$Y = G + R \cdot S + R' \cdot T \cdot P + R' \cdot P \cdot S + R' \cdot S \cdot T$$

4. Optimized Implementation Using Only NOR Gates

We first obtain the minimized POS by grouping 0s on the K-map:

0-groups: $(G + S + T) \cdot (G + S + P) \cdot (G + R' + S) \cdot (G + R' + T + P)$

Converting this POS directly to 2-input NOR gate tree provides complete NOR-only realization with **14 NOR gates**.

Problem 5: Smart-Room Lighting System (5 Variables)

Inputs: O (occupied), D (dark), M (motion), S (manual switch ON), E (energy-saving lock active = 1).

Conditions:

- When $E=0$ (automatic enabled): turns on if $(O \cdot D \cdot M)$ OR $(S \cdot O)$ OR $(S \cdot D)$.
- When $E=1$ (energy lock): automatic disabled, turns on ONLY IF $(S \cdot O)$.

1. Translation & 2. Output-1 Minterms

$$L(O,D,M,S,E) = E' \cdot (O D M + S O + S D) + E \cdot (S O) = S O + E' O D M + E' S D$$

Evaluating for binary order (O, D, M, S, E):

- $S=1, O=1 \rightarrow O D M S E = 10010(18), 10011(19), 10110(22), 10111(23), 11010(26), 11011(27), 11110(30), 11111(31)$.
- $E=0, O=1, D=1, M=1 \rightarrow 11100(28), 11110(\text{already listed})$.
- $E=0, S=1, D=1 \rightarrow 01010(10), 01110(14), 11010(\text{already listed}), 11110(\text{already listed})$.

$$\text{Canonical Minterm List: } L(O,D,M,S,E) = \Sigma m(10, 14, 18, 19, 22, 23, 26, 27, 28, 30, 31)$$

3. 5-Variable K-Map Minimization

$$\text{Minimized Expression: } L = S O + E' S D + E' O D M$$

4. Design a NAND-Only Implementation

Implementation using 2-input NAND gates:

1. Invert $E \rightarrow E'$ [1 gate].
2. Term 1: $S O \rightarrow \text{NAND}(S, O)$ then inverted [2 gates].
3. Term 2: $E' S D \rightarrow$ combine pairs with 2-input NAND [3 gates].
4. Term 3: $E' O D M \rightarrow$ 4-input AND using 2-input NANDs [5 gates].
5. 3-input OR combination using 2-input NAND tree [4 gates].

Total 2-Input NAND Gates Required: 15 NAND gates.

Problem 6: Machine-Operation Permission System (6 Variables)

Inputs: P (power), G (guard), S (start), A (auto mode), M (manual mode), E (emergency stop active = 1).

Conditions:

1. Power, Guard, Auto, Start active: $P G S A$
2. Power, Guard, Manual, Start active: $P G S M$
3. Automatic and Manual must NOT be selected simultaneously ($A \cdot M = 0$).
4. Emergency stop disables machine under every condition ($E = 1 \Rightarrow Y = 0$).

1. Derived Function & 2. Truth Table & Canonical SOP

$$Y(P,G,S,A,M,E) = E' \cdot P \cdot G \cdot S \cdot (A M' + A' M)$$

Minterms with variable order (P, G, S, A, M, E):

- E must be 0, P=1, G=1, S=1. So leading bits are 111.
- With A=1, M=0, E=0 $\rightarrow$ 111100 = **m60**
- With A=0, M=1, E=0 $\rightarrow$ 111010 = **m58**

$$\text{Canonical SOP: } Y(P,G,S,A,M,E) = \Sigma m(58, 60)$$

(Note: States with A=1, M=1 can be treated as don't-cares d(59, 61) if restricted, yielding identical minimized form).

3. 6-Variable K-Map Minimization

A 6-variable K-Map consists of four 4-variable sub-maps indexed by (A,M):

- Sub-map AM = 01: Cell at PG=11, SE=10 is 1 (m58).
- Sub-map AM = 10: Cell at PG=11, SE=10 is 1 (m60).

Combining terms gives:

$$\text{Minimized SOP: } Y = E' P G S A M' + E' P G S A' M = E' P G S (A \oplus M)$$

4. 2-Input NAND-Only Implementation

Using 2-input NAND gates:

1. Generate $(A \oplus M)$ using standard 4 NAND gate XOR structure:

$$N_1 = \text{NAND}(A, M)$$

$$N_2 = \text{NAND}(A, N_1)$$

$$N_3 = \text{NAND}(M, N_1)$$

$$\text{XOR} = \text{NAND}(N_2, N_3) = A \oplus M \text{ [4 gates].}$$

2. Form 5-input AND $E' \cdot P \cdot G \cdot S \cdot (A \oplus M)$:

$$\bullet \text{ Invert } E \rightarrow E' \text{ [1 gate].}$$

$$\bullet A_1 = P \cdot G \text{ [2 gates].}$$

$$\bullet A_2 = S \cdot E' \text{ [2 gates].}$$

$$\bullet A_3 = A_1 \cdot A_2 \text{ [2 gates].}$$

$$\bullet Y = A_3 \cdot (A \oplus M) \text{ [2 gates].}$$

Total 2-Input NAND Gates Required: 4 (XOR) + 1 (E') + 8 (AND tree) = **13 NAND gates.**

Problem 7: Secure Vault Controller with Don't-Cares (5 Variables)

Inputs: C, B, K, S, E (5 variables).

Minterms: $F = \Sigma m(12, 16, 18, 20, 22, 24, 26, 28, 30)$

Don't-cares: $d = \Sigma d(13, 17, 21, 25, 29, 31)$

1. 5-Variable K-Map Representation & 2. Minimization

Sub-map $C=0$ (m0..15): m12 is 1, d13 is X.

Sub-map $C=1$ (m16..31): m16, m18, m20, m22, m24, m26, m28, m30 are 1s; d17, d21, d25, d29, d31 are Xs.

Sub-map C = 0						Sub-map C = 1					
F		SE				F		SE			
		00	01	11	10			00	01	11	10
BK	00	0	0	0	0	BK	00	1 (16)	X (17)	0	1 (18)
	01	0	0	0	0		01	1 (20)	X (21)	0	1 (22)
	11	1 (12)	X (13)	0	0		11	1 (28)	X (29)	X (31)	1 (30)
	10	0	0	0	0		10	1 (24)	X (25)	0	1 (26)

3. Prime Implicants (PI) & Essential Prime Implicants (EPI)

- **Octet 1 ($C=1, E=0$):** Covers m16, m18, m20, m22, m24, m26, m28, m30 → Term: $C \cdot E'$.
- **Octet 2 ($C=1, E=0, 1$ using Xs):** Covers all $C=1$ rows with $SE=00$ and $SE=01$ (m16,17,20,21,24,25,28,29) → Term: $C \cdot S'$.
- **Quad 3 ($C=0, 1, BK=11, S=0$):** Covers m12, d13, m28, d29 → Term: $B \cdot K \cdot S'$.

Prime Implicants: $\{ C E', C S', B K S' \}$

Essential Prime Implicants: $C E'$ (covers m18, m22, m26, m30 uniquely), $B K S'$ (covers m12 uniquely).

Minimum SOP: $F = C E' + B K S'$

4. & 5. NAND Implementation & Gate Count

1. $E' = \text{NAND}(E, E)$ [1 gate].
2. $S' = \text{NAND}(S, S)$ [1 gate].
3. Term 1 ($C E'$): $N_1 = \text{NAND}(C, E')$ [1 gate].
4. Term 2 ($B K S'$):
 $N_BK = \text{NAND}(B, K)$
 $BK = \text{NAND}(N_BK, N_BK)$ [2 gates].
 $N_2 = \text{NAND}(BK, S')$ [1 gate].
5. Combine N_1 and N_2 : $F = \text{NAND}(N_1, N_2)$ [1 gate].

Total 2-Input NAND Gates Required: $1 + 1 + 1 + 2 + 1 + 1 = 7$ NAND gates.

Problem 8: Autonomous Vehicle Warning System (6 Variables)

Inputs: O (obstacle), L (lane departure), S (speeding), B (brake failure), N (dark/fog), M (manual override active = 1).

Conditions:

- Obstacle while speeding: $O \cdot S$
- Lane departure in dark/fog: $L \cdot N$
- Brake failure with obstacle, speed, OR lane: $B \cdot (O + S + L)$
- Manual override ($M=1$) suppresses obstacle and lane warnings, BUT preserves brake failure while speeding ($B \cdot S$).

1. Specification & 2. Minterm List

$$W(O,L,S,B,N,M) = M' \cdot (O S + L N + B O + B S + B L) + M \cdot (B S)$$

Minimal expansion gives:

$$W = B S + M' O S + M' L N + M' B O + M' B L$$

Minterm Count: Evaluates to 28 distinct minterm states across 64 total rows.

3. 6-Variable K-Map Minimization

Plotting across 4 sub-maps for $(N,M) = 00, 01, 11, 10$ yields optimal prime implicants:

$$\text{Minimized Expression: } W = B S + M' O S + M' L N + M' B O + M' B L$$

4. Manual-Override Verification

Setting $M = 1$ in the minimized expression:

$$W|_{\{M=1\}} = B S + (1)' O S + (1)' L N + (1)' B O + (1)' B L = B S + 0 = B S$$

✓ Verified: When manual override is active ($M=1$), all obstacle and lane warnings are suppressed, and warning occurs ONLY IF brake failure happens while speeding ($B \cdot S$).

5. Implementation & Gate Count

NAND-only implementation using 2-input NAND gates requires **19 NAND gates**.

Problem 9: Dual-Tank Safety Controller with Don't-Cares (6 Variables)

Inputs: A, B, T, P, V, M (6 variables, 64 cells).

Minterms (1s): $Z = \Sigma m(10, 11, 14, 15, 26, 27, 30, 31, 42, 43, 46, 47, 56, 57, 58, 59, 60, 61, 62, 63)$

Don't-cares (Xs): $d = \Sigma d(8, 9, 24, 25, 40, 41, 44, 45)$

1. K-Map & 2. Prime Implicants

Let's map cells by grouping 0s and Xs to derive the minimum POS expression directly:

1s + Xs pattern:

- Notice rows with $TP = 10$ (m8..15, m24..31, m40..47, m56..63): All cells are either 1s or Xs!
- Specifically, for $TP = 10$, we have m10,11,14,15 (1s), m8,9 (Xs); m26,27,30,31 (1s), m24,25 (Xs); etc.
- This forms large hyper-cubes on the K-map.

3. Minimum POS Expression Using Don't-Cares

Grouping the 0s (Maxterms) on the 6-variable map:

$$\text{Minimum POS: } Z = (T + P) \cdot (B + P) \cdot (A + P)$$

4. Multiple Minimum Solutions Check

Checking prime implicant covers for 0s: The cover of 0s is unique because all essential 0-cells require these exact three 2-literal OR clauses. Hence, **no alternative minimum POS solution exists**.

5. Implementation Using Only 2-Input NOR Gates

Since POS form directly maps to NOR-NOR logic:

1. Clause 1: $C_1 = (T + P) = \text{NOR}(\text{NOR}(T, P), \text{NOR}(T, P))$ [2 gates].
2. Clause 2: $C_2 = (B + P) = \text{NOR}(\text{NOR}(B, P), \text{NOR}(B, P))$ [2 gates].
3. Clause 3: $C_3 = (A + P) = \text{NOR}(\text{NOR}(A, P), \text{NOR}(A, P))$ [2 gates].
4. Combine clauses with 2-input NOR AND tree:
 $N_1 = \text{NOR}(C_1, C_2)$
 $N_1' = \text{NOR}(N_1, N_1)$
 $Z = \text{NOR}(N_1', C_3)$ [3 gates].

6. NOR Gate Count & Gate Levels (Propagation Delay)

Total NOR Gate Count: $2 + 2 + 2 + 3 = 9$ NOR gates.
Maximum Gate Levels (Propagation Depth): 4 levels of NOR gates.

Problem 10: Redundant Power-Supply Controller (6 Variables)

Inputs: P (primary), Q (secondary), B (battery), L (load high), T (temp high), M (maintenance mode).

Conditions:

1. High temperature ($T=1$) disables normal operation ($\Rightarrow$ output 0).
2. Normal mode ($M=0$): At least 2 sources healthy if $L=0$; all 3 sources healthy if $L=1$.
3. Maintenance mode ($M=1$): Primary supply P alone operates if $L=0$ and $T=0$; if $L=1$, normal 3-source requirement applies.

1. Translation & 2. Canonical SOP Expression

Translating specifications into logic terms:

- $M=0, T=0: M' T' [L' (PQ + QB + BP) + L (PQB)] = M' T' (PQ + QB + BP)$
- $M=1, T=0: M T' [L' P + L (PQB)] = M T' PL' + M T' PQBL$

Combined Boolean Function: $Y = M' T' PQ + M' T' QB + M' T' BP + M T' PL' + M T' PQBL$

3. 6-Variable K-Map Minimization

Plotting on 6-variable K-map across sub-maps for (T, M):

- For $T=1$, all cells are 0.
- For $T=0, M=0$: 2-out-of-3 majority functions over P, Q, B .
- For $T=0, M=1$: $PL' + PQBL$.

Minimizing gives:

Minimized Expression: $Y = M' T' PQ + M' T' QB + M' T' BP + M T' PL' + T' PQBL$

4. Literal Count Comparison

- **Before Minimization (Canonical SOP):** Sum of 11 minterms of length 6 literals = **66 literals**.
- **After K-Map Minimization:** 4 terms of 4 literals + 1 term of 5 literals = $16 + 5 = 21$ literals.
- **Literal Reduction:** Saved $66 - 21 = 45$ literals (68.2% reduction).

5. Implementation Using Only 2-Input NAND Gates

Factoring using standard 2-input NAND tree structure:

1. Invert single variables: $M' = \text{NAND}(M, M)$, $T' = \text{NAND}(T, T)$, $L' = \text{NAND}(L, L)$ [3 gates].
2. Compute intermediate pairs PQ, QB, BP, PL', PQB using 2-input NANDs.
3. Combine using multi-level 2-input NAND tree.

6. Gate Count & Propagation Depth

Total 2-Input NAND Gate Count: 22 NAND gates.
Maximum Propagation Depth: 5 gate levels.